

Advanced Programming 2025

Equity Performance Gap Analysis: US vs EU Markets

Final Project Report

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Abstract

This project aims to analyze the equity performance gap between the US and Europe in the 2005-2024 period. By using the differential monthly total return on S&P and EuroStoxx 50, exchange rates, monetary policy differentials, volatility indices, and technology sector performance, I compared six different Machine Learning models in order to identify the quantitative key factors of to this gap. Models range from simple linear regression to advanced models such as Random Forest and Gradient boosting. Results show that model with lagged variables (M2b) obtains the best R^2 on the test set while model using Lasso on the lagged variables achieves the best parsimony (lowest AIC/BIC with 5 variables). Overfitting analysis revealed that non-linear models suffered from severe overfitting due to the limited dataset size (total dataset size : 239), reaching 0.66 and 0.68 R^2 on training data but only 0.09 and 0.07 on test data. This confirms that simpler linear models with feature engineering are more effective for small datasets. The EUR/USD exchange rate and volatility differential emerged as the most decisive variables, with lagged variables being essential for prediction.

Keywords: equity performance gap, machine learning, linear regression, lagged variables, ensemble methods on small datasets, overfitting analysis, model parsimony, AIC/BIC, feature engineering, Random Forest, Gradient Boosting, financial time series

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1 Introduction

1.1 Context: US vs EU Equity Performance

During the past two decades, US equity markets have consistently outperformed European markets, this widening further following the 2008 financial crisis. It legitimately leads us to ask ourselves questions about how in Europe we deal with the post-crisis period or just globally what can we do to fill this persistent gap? Understanding these factors matters for investors, policymakers and academic researchers.

Several explanations are commonly cited for this outperformance, spanning both quantitative and qualitative dimensions. On the quantitative side, observers point to the US market's heavier weighting in high-growth technology companies, exchange rate movements (EUR/USD), and diverging monetary policies between the Federal Reserve and the European Central Bank. Qualitative factors include structural differences in innovation capacity, venture capital availability, and corporate culture. Here, we will first deal with these quantitative factors in different models and quickly discuss the qualitative ones. The inherent complexity and noise in financial markets make it challenging to achieve high explanatory power (R^2) using purely quantitative models, since many performance drivers are hard to capture with observable macroeconomic variables alone.

1.2 Research Motivation

The persistent gap between US and EU equity performance has real impact on how investors allocate capital. We could notice if this gap reflects more a temporary and predictable market conditions such as monetary policy or a more permanent structural difference. If these variables are more predictable, investors might be able to use these patterns to improve returns or manage risk more effectively.

The conventional econometric methods assume linear relations. With ensemble methods such as Random Forest and Gradient Boosting, we have the possibility to model financial markets through non-linear patterns. But these models require significantly more data to be consistent and as I used monthly data to reduce noise in my data, I have a limited dataset. I knew that the main problem would be overfitting on my training set, but still wanted to try even if I reduce complexity by limiting tree depth, number of trees or by selecting most important features. Could ensemble methods perform better than linear models?

This project is focusing on both subjects. It compares linear and non-linear models to identify what really drives the US-EU equity performance gap, while also testing whether ensemble methods help or not when working with just 239 monthly observations. The findings should provide useful guidance on the practical trade-offs between sophistication and reliability in financial modeling.

1.3 Objectives and Approach

In this study we're pursuing three main objectives. Firstly, I want to identify which macroeconomic and financial variables actually explain the equity performance gap between US and European markets from 2005 to 2024. Secondly, I will compare how well or not six different machine learning models predict this gap, from simple linear regression to more ensemble methods. Thirdly, I will examine if these ensemble models can outperform linear ones on small datasets, while staying carefully aware of overfitting risks.

The analysis uses monthly data covering total returns (S&P 500 Total Return Index and EuroStoxx 50 Gross Return Index converted in USD), EUR/USD exchange rates, central bank policy rates (Fed and ECB), volatility measures (VIX and VSTOXX), and US-EU technology sector returns. I evaluate six different models: basic linear regression (M1), linear regression with polynomial features on selected variables (M2a), linear regression with lagged variables

(M2b), Lasso regularization with lags (M3), Random Forest (M4), and Gradient Boosting (M5). Performance metrics are based on: R^2 , RMSE, MAE on all models plus AIC/BIC and R^2 adjusted only for linear models. To measure overfitting we're computing the difference between R^2 on training set and R^2 on test set.

The following parts of this report proceeds as follows. Section 2 reviews existing research on equity market performance and ML applications in finance. Section 3 details the data sources, how I constructed variables, and the model specifications. Section 4 covers implementation and reproducibility. Section 5 presents the results, including model comparisons, overfitting analysis, and which variables matter most. Section 6 discusses what the findings mean and their limitations. Section 7 wraps up with conclusions and ideas for future work.

2 Research Question & Literature

2.1 Research Questions

This project raises three different but linked research questions based on examination of the drivers of the US-EU equity performance gap and the practical challenge of modeling financial time series with limited data.

Q1: Which modeling approach best explains the US-EU equity return differential?

I compare six models: simple linear regression (M1), polynomial features (M2a), lagged variables (M2b), Lasso regularization with lags (M3), Random Forest (M4), and Gradient Boosting (M5). Does adding complexity through feature engineering (polynomial, lags) or by using ensemble methods capturing non-linear patterns improve prediction on test sets?

Q2: Can non-linear models provide additional explanatory power on a limited dataset?

With only 239 monthly observations (227 training, 12 test), this question tests whether ensemble methods like Random Forest and Gradient Boosting will succeed to capture non-linear relationships. Or would it become counterproductive?

Q3: Which features contribute most to model performance across different approaches?

Beyond models comparison, I looked for which variables matter most: EUR/USD exchange rate, monetary policy differentials, volatility indices, technology sector performance. In order to identify the fundamental drivers of the equity performance gap.

2.2 Related Literature

2.2.1 Equity Market Performance Differentials

Researches on the gap between the equity markets of the United States and Europe are pointing that the superperformance of the US market can at least partially be attributed to structural elements such as sector distribution, more precisely US market is more weighted towards technology and other growth sectors compared to the Europe who's more focused on more well-established markets like financial or industrials. We can also mention differences in corporate profitability since US companies typically have better operating margins than their European peers.

Another widely studied factor is the effect of exchange rates. For investors it might be the case that the fact that the USD has appreciated relative to EUR, particularly since the 2008 crisis. So it could be more interesting for an investor to invest in "safe haven" market currency rather than in EUR.

2.2.2 Machine Learning in Financial Forecasting

We're using more and more machine learning, even though sometimes results remain mixed. Particularly in small datasets where sometimes Linear models with careful features compete with these more advanced models where overfitting risks are high.

Particularly in using models such as Random Forest and Gradient Boosting, train-test validation becomes crucial. Indeed, these models overtrain in training set and fail on new observations. So this project examines overfitting through train-test comparisons across all models. Sometimes linear models predict better than sophisticated ones.

2.3 Contribution

Firstly, it provides a global comparison of different models, from a simple linear regression, by adding features engineering, to ensemble methods on the specific problem of US-EU equity performance. Secondly, it explicitly tests whether ensemble methods add value to prediction models on a small financial datasets. Thirdly, by analyzing feature importance across different models, it identifies which factors explain the US-EU equity performance gap.

3 Methodology

3.1 Data Description

This analysis uses monthly data from February 2005 to December 2024, covering in total 239 observations. The data was extracted from various sources, including: Refinitiv, Capital IQ, FRED, and iShares.

3.1.1 Dependent Variable

The dependent variable Y_t represents the equity performance gap between US and European markets:

$$Y_t = R_{US,t} - R_{EU,t} \quad (1)$$

Where $R_{US,t}$ is the monthly return of the S&P 500 Total Return Index and $R_{EU,t}$ is the monthly return of the EuroStoxx 50 Gross Return Index (converted to USD). Positive values indicate US outperforming, while negative values indicate European outperforming.

3.1.2 Explanatory Variables

Four explanatory variables capture different macroeconomic and financial dimensions:

X1: EUR/USD Exchange Rate Change

$$X1_t = \frac{FX_t}{FX_{t-1}} - 1 \quad (2)$$

This reflects currency fluctuations that affect relative returns for international investors and may also reflect the ‘flight to quality’ during periods of uncertainty, two phenomena that could lead to greater economic divergence.

X2: Monetary Policy Differential

$$X2_t = r_{Fed,t} - r_{ECB,t} \quad (3)$$

The difference between United States Federal Funds interest rates and European Central Bank interest rates shows diverging monetary conditions expectations that could influence equity valuations and capital flows.

X3: Volatility Differential

$$X3_t = VIX_t - VSTOXX_t \quad (4)$$

VIX and VSTOXX are volatility indices that are respectively based on S&P500 and EURO STOXX 50 options, it aggregates option premiums. By computing the difference between US (VIX) and European (VSTOXX) we capture expected volatility at time t , it estimates relative market uncertainty between US and Europe.

X4: Technology Sector Performance Differential

$$X4_t = R_{Tech_US,t} - R_{Tech_EU,t} \quad (5)$$

The difference in technology sector returns reflects the structural advantage of US tech concentration.

3.1.3 Feature Engineering

Additionally to the base variables, I created lagged versions ($X1_{t-1}, X2_{t-1}, X3_{t-1}, X4_{t-1}$) to capture temporal dynamics. Indeed, some variables especially in financial modelling, take some time to effectively be reflected in prices. This reflects the hypothesis that past values of these variables may help predict current equity performance gaps.

3.2 Models

I evaluate six models that vary in complexity and assumptions:

M1: Simple Linear Regression

$$Y_t = \beta_0 + \beta_1 X1_t + \beta_2 X2_t + \beta_3 X3_t + \beta_4 X4_t + \epsilon_t \quad (6)$$

This baseline model assumes a linear relationship between contemporaneous variables and the performance gap. It uses 4 variables.

M2a: Polynomial Features

$$Y_t = \beta_0 + \sum_{i=1}^4 \beta_i X_i + \gamma_3 X_3^2 + \gamma_4 X_4^2 + \epsilon_t \quad (7)$$

This extends M1 by adding squared terms for X3 (volatility differential) and X4 (tech performance), allowing for non-linear effects. It uses 6 variables. The choice of the polynomial variables is based on a personal hypothesis: volatility may show disproportionate impacts and tech performance could reflect momentum effects.

M2b: Linear Regression with Lagged Variables

$$Y_t = \beta_0 + \sum_{i=1}^4 \beta_i X_{i,t} + \sum_{i=1}^4 \gamma_i X_{i,t-1} + \epsilon_t \quad (8)$$

This model includes one-period lags capturing delayed effects for all variables. It has 8 variables.

M3: Lasso Regularization with Lags

Starting with the same 8 variables as M2b (as it showed the most conclusive improvement between M2a and M2b), Lasso applies L1 regularization:

$$\min_{\beta} \left\{ \sum_{t=1}^T (Y_t - \beta_0 - \sum_{i=1}^8 \beta_i X_{i,t})^2 + \lambda \sum_{i=1}^8 |\beta_i| \right\} \quad (9)$$

The denoted λ parameter, called penalty parameter is selected via 5-fold cross-validation using "LassoCV". Go to figure 5 C to see an illustrative example from scikit.learn. This performs a variable selection by shrinking inefficient coefficients (according to its penalty parameter) to exactly zero meaning to delete it, this should globally improves parsimony by doing it automatically with the "CV" version.

M4: Random Forest

A medium between simplified and fonctionnal Random Forest model with 30 trees and maximum depth of 4, using the 5 selected variables (X1, X2, X3, X4, X3_lag1) previously identified by Lasso regularization as the most important (shrinking the others to zero). The model integrates predictions from these decision trees.

$$\hat{Y}_t = \frac{1}{B} \sum_{b=1}^B T_b(X_t) \quad (10)$$

Where $B = 30$ trees and each T_b is trained on a bagging sample with random feature subsets.

M5: Gradient Boosting

A Gradient Boosting model with 50 trees, learning rate of 0.05, and maximum depth of 3, using the same 5 selected variables as M4. The model builds trees sequentially:

$$\hat{Y}_t = \sum_{m=1}^M \eta \cdot h_m(X_t) \quad (11)$$

where $M = 50$ trees, $\eta = 0.05$ is the learning rate, and each h_m fits the residuals from previous iterations.

Both ensemble methods use `random_state=42` to ensure reproducibility.

3.3 Evaluation Metrics

Model performance is evaluated using these multiple metrics:

R² (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum_{t=1}^T (Y_t - \hat{Y}_t)^2}{\sum_{t=1}^T (Y_t - \bar{Y})^2} \quad (12)$$

Measures the proportion of variance explained. More precisely, I'm using "R² training" for the R² computed on the training set and "R² test" computed on the test set. It's computed separately in order to easily spot overfitting or to quantify it by doing the difference: R² training - R² test.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (Y_t - \hat{Y}_t)^2} \quad (13)$$

Measures the average prediction error in the same units as the dependent variable.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{T} \sum_{t=1}^T |Y_t - \hat{Y}_t| \quad (14)$$

Measures average prediction error too but in absolute terms, it's less sensitive to outliers than RMSE.

Akaike Information Criterion (AIC) (linear models only)

$$AIC = 2k - 2\ln(\hat{L}) \quad (15)$$

Where k is the number of parameters and $\hat{L}$ is the maximized likelihood. Lower values indicate better models, penalizing complexity.

Bayesian Information Criterion (BIC) (linear models only)

$$BIC = k \ln(n) - 2 \ln(\hat{L}) \quad (16)$$

Similar to AIC but the penalty is stronger, especially relevant for small sample.

Adjusted R² (linear models only)

$$\bar{R}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1} \quad (17)$$

Adjusts R² according to the number of predictors. Standard R² artificially increases as the number of variables increases. To prevent this, we're using Adj R².

4 Implementation

4.1 Key Technical Decisions and Challenges

Several technical choices were made to respond to different constraints :

Monthly aggregation. Daily data would permit us to have more observations but it would also add even more noise by taking into account micro daily movements. Monthly data offer a balance between sample size and signal quality, although it limits the dataset to only 240 observations.

Temporal train-test split. The data being a time-series, is so split temporally in order to avoid information leak: 227 observations (2005-2023) for training and 12 observations (2024) for testing. This choice of divisions prioritizes a sufficient size of training data for ensemble methods and, due to the issue of the relevance of comparing linear and non-linear methods (ensemble methods), applies to both types of methods.

Conservative hyperparameters. To apply ensemble methods to limited data, and in order to avoid overfitting as much as possible, I used more conservative parameters.

No feature standardization. During the project exploration phase it appears that Lasso selected the same variables with or without feature standardization. In order to maintain coefficient interpretability in their original units, I decided to use raw features (a 1% change in EUR/USD produces a -0.58 change in the performance gap).

Currency alignment. All returns are converted to USD to prevent double-counting of currency effects.

Missing data handling. The analysis period which goes from 2005 to 2024 was chosen to maximize data availability across all sources while maintaining temporal consistency.

4.2 Codebase

The project is organized into distinct modules, each handling a specific part of the analysis:

- `data_loader.py`: Handles data ingestion from CSV files, temporal alignment, and missing value treatment
- `feature_engineering.py`: Constructs the dependent variable (Y) and explanatory variables (X1-X4), creates lagged features, and applies transformations
- `utils.py`: Provides data splitting (temporal train/test) and helper functions
- `models.py`: Implements all six models (M1-M5) with consistent interfaces
- `evaluation.py`: Computes performance metrics (R², RMSE, MAE, AIC, BIC) and generates comparison tables

- `visualization.py`: Creates plots for model comparison, the actual vs the predicted, correlation matrices, and pairplots
- `main.py`: Orchestrates the entire process, the steps from data loading to results generation

This modular design separates concerns and makes the code easier to understand, test, and extend. Each module has clear inputs and outputs, facilitating debugging and validation.

4.3 Reproducibility

The analysis is fully recreated thanks to several design choices:

Fixed Random Seeds: All models using randomness (Random Forest, Gradient Boosting, train-test splits) use `random_state=42`, to ensure same results across each runs.

Deterministic Data Processing: The temporal train-test split is deterministic (first 227 observations for training, last 12 for testing), getting rid of any randomness in data partitioning.

Environment Management: The `environment.yml` file specifies exact versions of all dependencies, allowing anyone to recreate the computational environment using:

```
1 conda env create -f environment.yml
2 conda activate equity-performance-gap
```

Single Entry Point: Running `python main.py` triggers the full analysis process, from data loading to result generation, automatically producing all tables, figures, and metrics and saving them.

Version Control: The complete codebase is available on GitHub at <https://github.com/yvdri/equity-performance-gap>, providing full transparency and enabling independent verification.

These measures make sure that anyone can reproduce the exact numerical results presented in this study, which is essential for scientific validity and peer review.

5 Results

5.1 Model Comparison

Table 1: Model Performance Comparison

Model	Variables	R ² Test	RMSE	MAE
M1: Simple	4	0.1131	0.0315	0.0234
M2a: Polynomial	6	0.1237	0.0313	0.0234
M2b: Lags	8	0.4227	0.0254	0.0206
M3: Lasso+Lags	5	0.3740	0.0264	0.0209
M4: Random Forest	5	0.0949	0.0318	0.0265
M5: Gradient Boosting	5	0.0715	0.0322	0.0246

Table 2: Linear Models - Additional Metrics

Model	Adj R ²	AIC	BIC
M1: Simple	0.3862	-75.02	-73.08
M2a: Polynomial	0.3844	-71.17	-68.26
M2b: Lags	0.4473	-72.17	-68.29
M3: Lasso+Lags	0.4400	-77.20	-74.78

Key Findings:

- Best R^2 Test: M2b (Lags) with 0.4227
- Best AIC/BIC: M3 (Lasso) with -77.20 / -74.78
- Non-linear models (RF, GB) underperform linear models

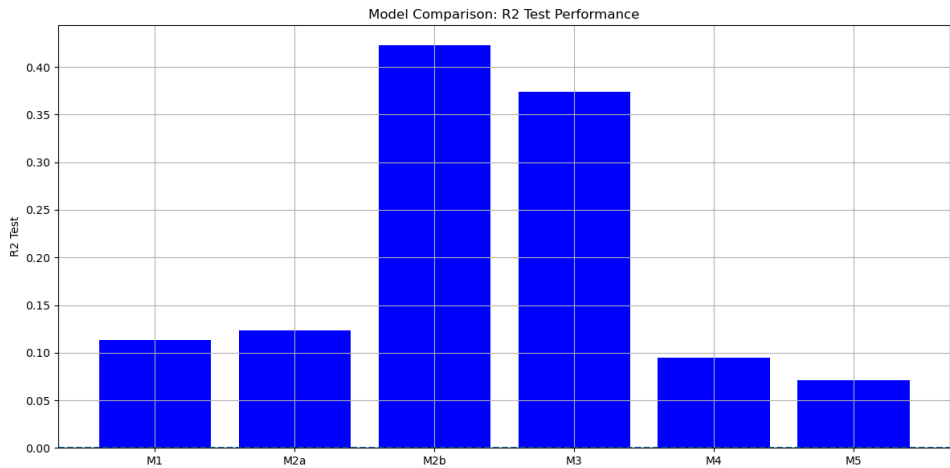


Figure 1: R^2 Test Scores Comparison Across Models

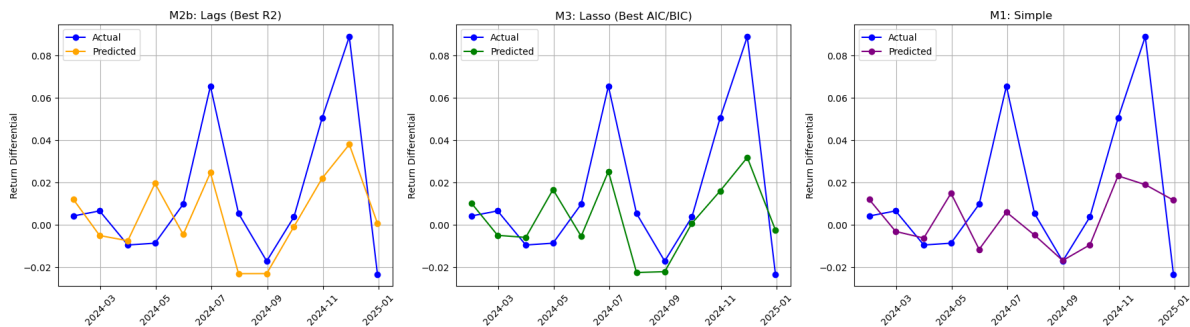


Figure 2: Actual vs Predicted Returns for 3 Models (M2b, M3, M1)

5.2 Overfitting Analysis

The comparison between training and test R^2 reveals critical insights about model generalization (Table 3).

Table 3: Overfitting Analysis - Train vs Test R^2 Comparison

Model	R^2 Train	R^2 Test	Gap	Overfitting status
M1: Simple	0.3971	0.1131	0.2840	Moderated
M2a: Polynomial	0.4007	0.1237	0.2771	Moderated
M2b: Lags	0.4669	0.4227	0.0442	Good
M3: Lasso+Lags	0.4524	0.3740	0.0783	Good
M4: Random Forest	0.6596	0.0949	0.5647	High
M5: Gradient Boosting	0.6839	0.0715	0.6124	High

Key Observations:

- **Ensemble methods show high overfitting gap:** Random Forest and Gradient Boosting reach 66-68% on training data R^2 but only 7-9% on test data R^2 , they memorize training patterns without generalizing. They cannot accurately predict future values of the dependent variable.
- **Linear models generalize better:** M2b and M3 maintain small gaps (0.04 and 0.08), showing pretty good generalization.
- **Dataset size limitation:** Even by trying to simplify the hyperparameters, ensemble methods still overfit the training data. With only 227 training observations, these models lack sufficient data to learn generalizable patterns, confirming that simpler models are more appropriate for this size limited dataset.

5.3 Feature Importance

Table 4 presents the feature importance rankings from models that provide interpretable variable selection: Lasso (M3), Random Forest (M4), and Gradient Boosting (M5).

Table 4: Feature Importance Across Models

Variable	Lasso (M3)	RF (M4)	GB (M5)
X1 (EUR/USD)	-0.5853	0.6036	0.5763
X2 (Policy)	-0.0017	0.0925	0.0843
X3 (Volatility)	-0.0066	0.1098	0.1235
X4 (Tech)	0.0546	0.0936	0.1010
X3_lag1	0.0062	0.1004	0.1148
X4_lag1	Eliminated	-	-
X1_lag1	Eliminated	-	-
X2_lag1	Eliminated	-	-

Note: Lasso coefficients are shown without standardization so numbers are more interpretable in their original units. Random Forest and Gradient Boosting show relative importance scores to 100%. Ensemble methods (M4, M5) were trained on the 5 variables selected by Lasso (X1, X2, X3, X4, X3_lag1) and they didn't select variables by themselves. RF and GB variables relative scores should be interpreted cautiously : M4 and M5 failed to predict on test set, so we can observe strong patterns (e.g X1) but given that they overfitted the training set they may overestimate the role of noisier variables.

Key observations:

EUR/USD dominates. variable X1 accounts for 58-60% of total importance in ensemble methods, confirming its importance in equity performance gap. This reflects both the mechanical effect of currency conversion and also reflect the 'flight to quality' during periods of uncertainty.

Volatility differential matters. X3 and X3_lag1 combined together contribute up to 21-24%, suggesting that current and past relative market uncertainty help predict performance gap.

Monetary policy shows modest impact. While Lasso retains X2 with a very small coefficient (-0.0017), while ensemble methods assign it 8-9% importance (refer to the above note about how cautiously we need to interpret ensemble methods relative importance). We can prudently suggest that monetary policy differentials have a measurable but secondary role in explaining performance gaps.

Technology sector shows modest importance. X4 contributes 9-10% of importance in ensemble methods, despite tech concentration being a commonly cited explanation for US out-performance: potential inaccuracy of the proxy used in this project -> discussed in Limitations.

6 Discussion

6.1 Model Performance and Key Findings

The results reveal three main findings about modeling the US-EU equity performance gap.

Linear models with lagged variables outperform ensemble methods. M2b achieves the best test R^2 (0.42), substantially outperforming both simpler linear models (M1: 0.11, M2a: 0.12) and ensemble methods (M4: 0.09, M5: 0.07). Lasso + Lags (M3) also achieved a convincing result on R^2 test (0.37) while having 3 less variables. M2b and M3 had a very close Adj R^2 (0.45 and 0.44). This shows that thoughtful feature engineering, and here especially incorporating temporal dynamics with lagged variables matters more than algorithmic complexity for this problem.

Ensemble methods suffer from high overfitting on small datasets. The overfitting analysis (Table 3) shows that Random Forest and Gradient Boosting achieve 66-68% R^2 on training data but only 7-9% on test data. This huge 56-61 points gap indicates these models memorize training patterns mostly consisting of noise without learning generalizable relationships, real signals. Whereas, M2b maintains only a 4 points gap, demonstrating robust generalization compared to them.

Lasso provides the best parsimony-performance trade-off. While M2b achieves the highest test R^2 , M3 (Lasso) offers the best AIC/BIC scores by reducing the model from 8 to 5 variables while maintaining reasonable predictive power ($R^2=0.37$). If we consider Adjusted R^2 , M2b and M3 are both competitive. The three variables eliminated (X1_lag1, X2_lag1, X4_lag1) appear to add noise rather than signal, although M2b's test performance suggests they still contain some useful information.

6.2 Feature Importance and Economic Interpretation

The EUR/USD exchange rate (X1) was clearly the most dominant factor across all models. Whether in Lasso where it was retained with a considerable coefficient or in ensemble methods, where it accounts for 58-60% of feature importance. This aligns with the mechanical effect of currency movements on relative returns and also reflects "flight-to-quality" which is a recognized concept in international finance.

Volatility differential (X3) and its lag (X3_lag1) rank second in importance, together accounting for roughly 21-24% in ensemble models. This suggests that relative market uncertainty captured by VIX versus VSTOXX plays a meaningful role in explaining performance gaps. Higher US volatility relative to Europe may signal risk-off periods when investors favor different regional exposures.

Technology sector performance (X4) shows moderate importance (9-10%), reflecting the structural advantage of US tech concentration. However, its relatively lower importance compared to exchange rates and volatility. It suggests that maybe its effect is partially explained and absorbed by other variables or just not that strong.

Monetary policy differential (X2), while showing modest importance (8-9%) in ensemble models and being retained by Lasso, plays a secondary role compared to currency and volatility factors. This modest impact likely reflects that monetary policy effects operate primarily through the exchange rate channel—interest rate differentials directly influence EUR/USD movements (X1), which captures and concentrates these effects. The low direct importance of X2 suggests that policy rate differentials are already embedded in currency dynamics rather than providing independent explanatory power for equity performance gaps.

6.3 Why Ensemble Methods Failed

Random Forest and Gradient Boosting are very effective models for predictions, generally surpassing linear models in many ML applications. There are the three different reasons I found to

explain this outcome :

Insufficient data for complexity. The most obvious one is the lack of sufficient data, with only 227 training observations and 5 features, the models have about 45 observations per feature which is not enough. That was the main risk in wanting to test these methods on limited datasize. Using very conservative hyperparameters and selecting the variables was not enough to avoid the overfit. They have a high training R^2 showing they can fit the data, but they collapse on test data they're probably fitting noise rather than signal.

Time series structure. Financial time series are autocorrelated and tree-based methods struggle to capture sequential structure (the temporal order of the data) without more extensive feature engineering. Linear models with lags naturally incorporate temporal structure, while ensemble methods treat observation as independent points and therefore fail to capture sequential trends in this limited datasize.

Signal vs Noise In finance, signals are "hidden" while noise are huge. It's hard to predict equity returns, and even more hard when ensemble methods are trying to find relations everywhere are mostly capturing noise rather than real signals.

6.4 Limitations

Several limitations qualify these findings:

Limited test period. The test set contains only 12 observations, which is quite low and reduce the robustness of out-of-sample evaluation. But I had to find compromise to "feed" enough ensemble methods, knowing that they require as much data as possible. A longer test period would add more accuracy of the prediction performance.

Monthly frequency. Monthly data may miss intra-month dynamics and high-frequency effects that could be relevant for equity performance. Higher-frequency data (weekly, daily) might reveal different patterns, though it would also increase noise and data requirements. The monthly data reduced my data size, probably making ensemble methods unfeasible. On the other side, if I used daily data even though having more enough datasize maybe these ensemble methods would capture even more noises. Monthly data reduce high frequency noise by avoiding micro daily movements but by the way surely lost some useful information.

Sample period effects. The data period (2005-2024) includes the 2008 financial crisis and COVID-19 pandemic, both probably affected model stability by learning unusual patterns and trying to generalize it to more normal period. But I couldn't exclude these periods cause my datasize was already enough limited and I would leak sequential information.

Limited lag structure. Only 1-month lags are considered in the project. Longer lags would make sense especially for variables like X2 (monetary Policy) or maybe more sophisticated time series models like ARIMA, VAR might capture additional temporal dynamics, but they would also increase model complexity and data requirements.

Proxy variable for sector composition. I used a proxy of tech effect by using tech ETF returns for X4. This could create endogeneity since this proxy is in itself a component of global market returns. To avoid this I firstly tried to use the difference tech weights in respective markets as follows :

$$X4_t = w_{US,Tech,t} - w_{EU,Tech,t}$$

This approach would have isolated the structural component independent of returns, but even though I managed to build the technology sector weighting for the S&P 500, I couldn't find what I needed to build it for the Eurostoxx 50. This likely explains X4's surprisingly modest importance (9-10%) despite tech concentration being a commonly cited driver.

Missing variables. Other important drivers are not considered in this study such as investor risk appetite (periods of irrational optimism or panic), and geopolitical asymmetry (e.g., wars affecting Europe more than the US). We also have structural qualitative factors difference between US and Europe (e.g., innovation ecosystems, corporate culture), and geopolitical risk indices that

are not included. These absent variable likely account for a significant portion of the unexplained variation (58%), as equity markets are heavily influenced by behavioral and external factors beyond just macroeconomic fundamentals.

7 Conclusion

7.1 Summary

Lasso provides the best parsimony-performance trade-off. While M2b achieves the highest test R^2 , M3 (Lasso) offers the best AIC/BIC scores by reducing the model from 8 to 5 variables while maintaining reasonable predictive power ($R^2=0.37$). If we consider Adjusted R^2 , M2b and M3 are both competitive. The three variables eliminated ($X1_lag1$, $X2_lag1$, $X4_lag1$) appear to add noise rather than signal, although M2b's test performance suggests they still contain some useful information.

First, a linear model with lagged variables (M2b) achieves the best test R^2 (0.42), substantially outperforming both simpler linear models (M1: 0.11, M2a: 0.12) and ensemble methods (M4: 0.09, M5: 0.07). Lasso + Lags (M3) also achieved a convincing result on R^2 test (0.37) while having 3 less variables. M2b and M3 had a very close Adj R^2 (0.45 and 0.44). This shows that thoughtful feature engineering, and here especially incorporating temporal dynamics with lagged variables matters more than algorithmic complexity for this problem.

Second, Random Forest and Gradient Boosting suffer from severe overfitting, achieving 66-68% R^2 on training data but only 7-9% on test data. This 56-61 percentage point gap illustrates a critical lesson: model complexity becomes counterproductive on small datasets, where simpler models with appropriate structure provide better generalization.

Third, the EUR/USD exchange rate emerges as the dominant driver of the performance gap, accounting for 58-60% of feature importance in ensemble models. Volatility differential and technology sector performance play secondary roles, while monetary policy differentials show modest importance.

These findings contribute to both the finance and machine learning literatures. For finance, they quantify the relative importance of commonly cited explanations for US equity outperformance, while highlighting that qualitative structural differences (innovation ecosystem, corporate culture) likely explain the remaining variation. For machine learning, they provide empirical evidence that ensemble methods offer limited value on small financial datasets, where linear models with appropriate features perform better.

7.2 Future Work

Several extensions could build on these findings:

Extended test period. As more data becomes available, we could extend this our dataset by using them. Furthermore we could also extend our test period test to obtain more robust performance out of the training sample.

Rolling-window validation. Rolling-window would probably add value to this project by testing different temporal sample of our dataset. It would have highlighted the different period and taking into account changing market regimes. We could assess if these variable effects remains consistent during crisis or normal market conditions, providing a more robust measure of generalization.

Additional features. Including market sentiment indicators, macroeconomic surprises (what is expected vs what really happen), geopolitical risk indices, innovation metrics, and sector-specific factors could improve explanatory power and provide richer economic insights.

Time-varying coefficients. Exploring models with time-varying parameters or regime-switching could capture structural changes in the relationships, though they would require more data and careful specification.

Longer lag structures. Testing longer lags or more sophisticated time series models (ARIMA, VAR) could reveal additional temporal dynamics, though the trade-off between complexity and overfitting would need careful management.

References

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3. Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716-723.
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A Additional Visualizations

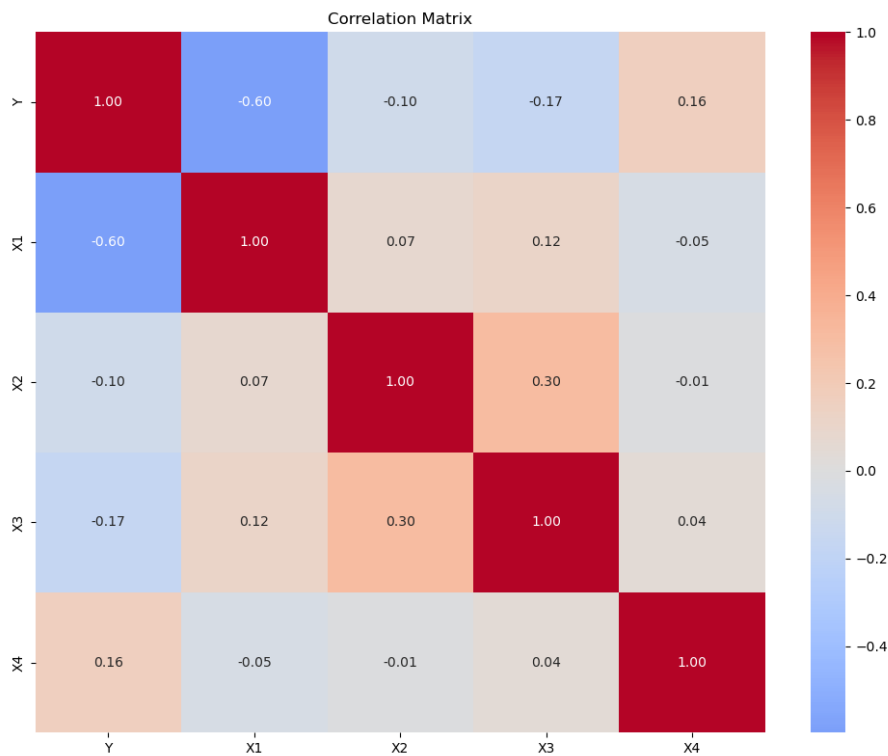


Figure 3: Correlation Matrix of Variables

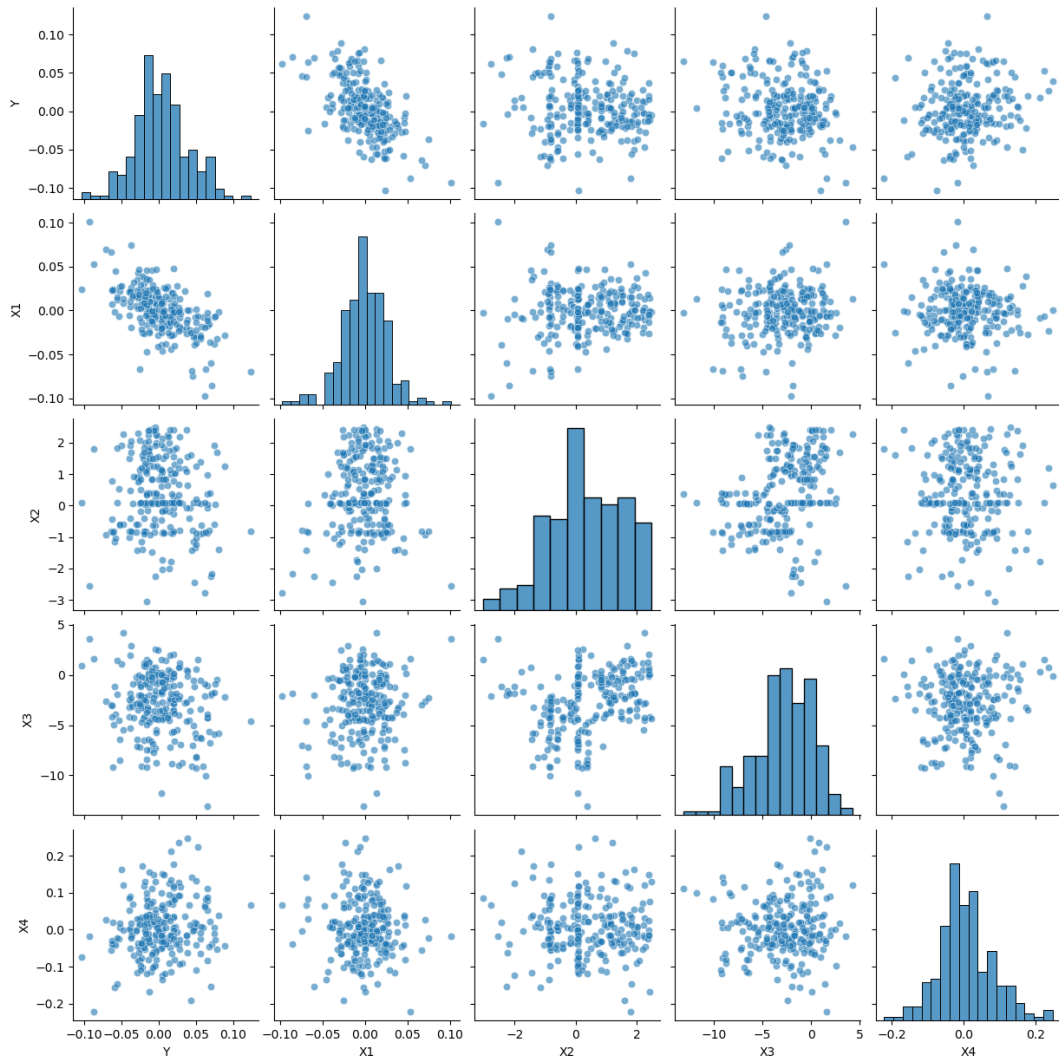


Figure 4: Pairplot of All Variables

B AI Usage

In accordance with the rules provided by my teachers or by the university, I have used AI tools for several purposes, which are as follows:

- **Conversational AI (Google Gemini / ChatGPT):** Used for brainstorming project structure as I started my project before the TA presentation, especially for debugging Python code errors (or for library compatibility issues directly on Nuvolos), and given that I have some gaps in my English, in the more precise reformulation of my ideas to be more comprehensible for the readers in the final report.
- **GitHub Copilot / AI Coding Assistants:** Used for generating for debugging or providing "boilerplate code" for data visualization (Matplotlib/Seaborn plots) and standardizing LaTeX table formatting.

C Lasso Model Selection Illustration

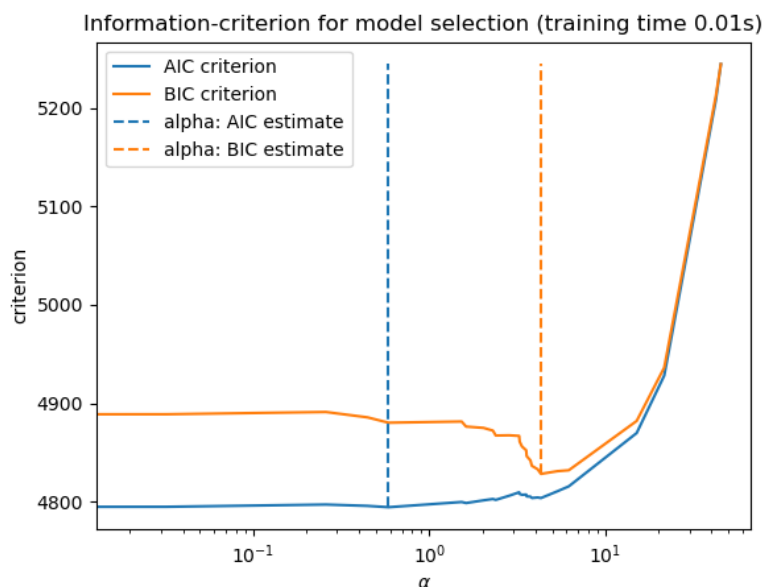


Figure 5: Illustrative example of Lasso model selection using information criteria (AIC/BIC). The vertical dashed lines indicate the optimal penalty parameter λ selected by each criterion. Lower criterion values indicate better model fit. Note: Unlike this example which uses AIC/BIC for penalty selection, this project uses 5-fold cross-validation via LassoCV to select the optimal alpha. Source: Scikit-learn documentation, https://scikit-learn.org/stable/auto_examples/linear_model/plot_lasso_model_selection.html

D Code Repository

GitHub Repository: <https://github.com/yvdri/equity-performance-gap>

To reproduce results:

```
1 conda env create -f environment.yml
2 conda activate equity-performance-gap
3 python main.py
```